Book 1. Stimulus-Only Analyses

What was actually in the auditory superstitious-perception stimulus set?

Project notebook

April 2026

Contents

Executive summary **2**

How to read this book **2**

1 Foundations (Part I) **3**
- 1.1 Stimulus inventory . . . 3
- 1.2 Preprocessing . . . 3
- 1.3 Descriptive properties . . . 3

2 Pearson-centered analyses (Part II) **3**
- 2.1 Chapter 4: Template-correlation distribution . . . 3
- 2.2 Chapter 5: Pairwise Pearson structure . . . 4
- 2.3 Chapter 6: Target cohesion vs. selection rank . . . 5
- 2.4 Chapter 7: Template alignment vs. pairwise cohesion . . . 6
- 2.5 Chapter 8: Pearson-based geometry of the full set . . . 7
- 2.6 Chapter 9: Template projection and residual structure . . . 8
  - 2.6.1 Decomposition . . . 8
  - 2.6.2 Mean contributions . . . 9
  - 2.6.3 Per-sample contributions (Chapter 9b) . . . 9
- 2.7 Chapter 10: Null models for Pearson structure . . . 10

3 Non-Pearson metric analyses (Part III) **11**
- 3.1 Chapter 11: Choosing alternative metrics . . . 11
- 3.2 Chapter 12: Label-aware separation under each metric . . . 12
- 3.3 Chapter 13: Label-blind clustering under each metric . . . 13
- 3.4 Chapter 14: Cross-metric synthesis . . . 13

4 Synthesis and behavioural cross-check (Part IV) **14**
- 4.1 Chapter 15: What did subjects do? . . . 14

5 What the stimulus set actually contains **16**

A File map **17**

B Key formulae cheat sheet **17**

Executive summary

The stimulus set contains **300 white-noise clips**, each $L = 2960$ samples long at 8 kHz (0.37 s), divided by design into 150 *targets* (highly correlated with a fixed template clip `wall.wav`) and 150 *distractors* (uncorrelated with it). Book 1 asks: *independent of any subject behaviour, what structure does this set actually contain?*

The short answer:

- The target/distractor split *is* statistically real under Pearson ($d = 0.31$, label-permutation $p < 10^{-3}$), but the effect size is tiny: targets are pairwise more similar than distractors by $\Delta\bar{r} \approx 5.8 \times 10^{-3}$.
- That entire effect is the projection onto the template `wall.wav`. After residualising against the template, target–target cohesion *vanishes* ($\bar{r}_{tt} = 3.5 \times 10^{-5}$, $p \approx 0.84$). The targets do not form an internal family beyond their shared wall-component.
- Under non-Pearson metrics (lagged cross-correlation, envelope correlation, log-spectrum correlation, MFCC cosine) the target/distractor distinction *collapses* ($d \leq 0.04$ for all four). The label is a Pearson-specific artefact of selection.
- The set has no clean intrinsic clusters: best silhouette across all metrics and clustering methods is 0.15 (envelope, agglomerative), and that split isolates 6 acoustic outliers from the other 294.
- Behaviourally, 26 subjects classified at chance (mean accuracy 0.510, hit rate 0.480, false-alarm rate 0.460). No metric predicts trial responses meaningfully (best AUC = 0.510). Label-blind clusters fit subjects' choices fractionally *better* than the experimenter labels do (0.530 vs. 0.510).

The detailed development follows.

How to read this book

The four parts walk a single arc:

- **Part I (Foundations).** What is on disk: 300 short white-noise clips, half labelled *target* (high Pearson correlation with a single fixed template clip `wall.wav`), half labelled *distractor* (near-zero correlation). Everything else in the book is built on these centred waveforms.
- **Part II (Pearson).** We compare the targets against the distractors using the same metric the experimenter used to build the set in the first place (Pearson correlation between centred waveforms). This is the metric most "flattering" to the design, and even here the effect is small.
- **Part III (other acoustic metrics).** Four other reasonable perceptual / acoustic similarity measures (lagged cross-correlation, envelope correlation, log-spectrum correlation, MFCC cosine). Under these, the target / distractor distinction collapses — the label is a Pearson-specific selection artefact.
- **Part IV (synthesis + behavioural sanity check).** A one-pass check against subject responses (full Book 2 follow-up is in `book2_writeup.pdf`) showing that subjects performed at chance and that no acoustic metric meaningfully predicts their responses.

Each chapter starts with a one-paragraph *Plain-English intent* box, then gives the formula and the table or plot, and ends with a one-paragraph *Reader takeaway* box. Skim only the boxes for the narrative; read the body for derivations and exact numbers.

1 Foundations (Part I)

1.1 Stimulus inventory

The stimulus set is organised on disk as

`raw_data/audio_stimuli/{full_sentence,imagined_sentence}/{targets,distractors}/`

giving 4 folders of 75 `.wav` files each, for 300 total. The folder split corresponds to the two *blocks* of the experiment: two folders per block, one for targets and one for distractors.

The selection rule for targets and distractors was Pearson correlation with a single fixed template, `raw_data/audio_stimuli/wall.wav`. Concretely, the original pipeline drew $\sim$ 2,000,000 candidate white-noise clips, computed each candidate's Pearson r to `wall.wav`, and selected the top tail as targets and a band near $r = 0$ as distractors.

1.2 Preprocessing

The on-disk files are PCM_16, 8 kHz, RMS-normalised to -23 dBFS as the final step of the pipeline. All Book 1 analyses load these on-disk samples directly as `float64` and *centre* them (subtract the mean) but do *not* z-score:

Listing 1: Canonical loader (excerpt from `ch11-non-pearson-metrics.py`)

```
sr, raw = wavfile.read(wav_path)
rows.append(raw.astype(np.float64))
...
waveform_matrix = np.vstack(rows)
waveform_matrix = waveform_matrix - waveform_matrix.mean(axis = 1, keepdims = True)
```

This matters because Pearson correlation is defined on centred signals but is invariant to scale, and because the energy-based metrics in Part III *depend* on absolute amplitude. Z-scoring would erase those amplitude differences.

1.3 Descriptive properties

Every clip has the same length ($L = 2960$ samples), the same sample rate (8000 Hz) and the same nominal RMS. Mean off-diagonal Pearson similarity across all 300×300 pairs is $+1.40 \times 10^{-3}$ with std 1.86×10^{-2}: the clips look like nearly orthogonal noise vectors in $\mathbb{R}^{2960}$.

2 Pearson-centered analyses (Part II)

2.1 Chapter 4: Template-correlation distribution

Plain-English intent: Targets were selected by sorting candidate clips on Pearson correlation with one fixed template, `wall.wav`. We start by simply looking at the distribution of those template correlations, both to confirm the selection rule and to set the scale (in $\mathbb{R}^{2960}$, what *counts* as a large correlation?).

For a centred clip $s \in \mathbb{R}^L$ and centred template $t \in \mathbb{R}^L$, Pearson r is

$$r(s,t) \;=\; \frac{\langle s,t\rangle}{\|s\|\,\|t\|} \;=\; \cos\theta_{st}. \tag{1}$$

Selected targets sit in the top tail of $r(\cdot, t_{\text{wall}})$ over the candidate pool; selected distractors sit near 0. In the final set we have $r(s, t_{\text{wall}}) \in [0.070,\, 0.094]$ for targets and $|r| \lesssim 10^{-2}$ for distractors (these are the values stored in `ch9_pearson_r_to_template.csv`).

The numerical smallness of even the targets ($r \approx 0.08$, not 0.5) is a consequence of the high dimension: a uniformly random unit vector in $\mathbb{R}^{2960}$ has r to a fixed unit vector with std $1/\sqrt{L} \approx 0.0184$, which matches the empirical distractor-distractor std almost exactly. Targets sit roughly 4–5σ out in that null distribution.

Reader takeaway: Targets really are extreme on the template axis ($\sim 4\sigma$), but the absolute correlations involved are tiny ($r \approx 0.08$). High dimension makes *small numerical correlations into large statistical distinctions*; keep that in mind whenever you see $r \sim 10^{-3}$ later.

2.2 Chapter 5: Pairwise Pearson structure

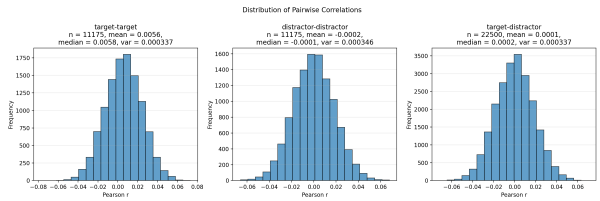
Plain-English intent: If targets share something acoustic, target–target pairs should be more Pearson-correlated than distractor–distractor or target–distractor pairs. We compute all three pairwise distributions and compare them.

Compute the full 300×300 correlation matrix and split the upper triangle into three pools by label: *target–target* ($n = 11{,}175$), *distractor–distractor* ($n = 11{,}175$), *target–distractor* ($n = 22{,}500$).

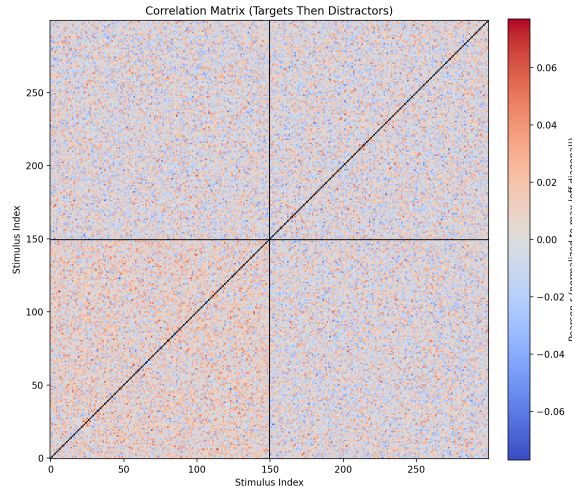
Table 1: Pairwise Pearson distributions by group.

Pool	mean	std	vs. zero (t, p)	vs. *dd* (Welch p)
target–target	+0.00561	0.0183	32.3, 2×10^{-219}	1.7×10^{-119}
distractor–distractor	−0.00017	0.0186	−0.95, 0.34	—
target–distractor	+0.00008	0.0184	0.66, 0.51	—

Effect sizes: $d_{tt-dd} = 0.313$, $d_{tt-td} = 0.301$, $d_{dd-td} = -0.013$. In words: *targets* pairwise correlate slightly above zero; distractors and cross-pairs are statistically indistinguishable from zero.



(a) Three pairwise distributions overlaid.



(b) 300×300 correlation matrix, ordered targets-then-distractors. The faint warm block in the bottom-left is the entire effect.

Figure 1: Pairwise Pearson structure (Chapter 5).

Reader takeaway: There *is* a target–target excess ($\bar{r}_{tt} \approx +0.0056$ vs. -0.0002 for distractors), Cohen's $d \approx 0.31$, and it is statistically air-tight ($p \sim 10^{-219}$). But the magnitude is the size of the noise std (both ~ 0.018). Targets are not a tight family; they have a shared mean offset of order 1%.

2.3 Chapter 6: Target cohesion vs. selection rank

Plain-English intent: If the cohesion in chapter 5 reflects the targets containing more of the template, the targets that are *most* like the template should be more cohesive among themselves. Restrict to the top-k most-template-like targets and recompute mean within-target Pearson, sweeping k.

Order targets by their template correlation $r(s_i, t_{\text{wall}})$, take the top k for $k \in \{10, 20, \ldots, 150\}$, and compute mean within-pool pairwise Pearson. If extreme-template targets formed a tighter family, $\bar{r}_{tt}^{(k)}$ should rise as k shrinks.

Table 2: Mean within-target pairwise r for top-k subsets.

k	$\bar{r}_{tt}^{(k)}$	SE
10	0.01189	0.00275
20	0.00671	0.00131
50	0.00618	0.00051
100	0.00565	0.00026
150	0.00561	0.00017

There is a small but real top-tail bump (top 10: $\bar{r} \approx 0.012$, roughly double the full-set mean), then immediate convergence to the full-set value by $k \geq 30$. Selection extremity drives a few extra percentage points of cohesion in the very top, no more.

Reader takeaway: Top-10 targets cohere about twice as strongly as the average target ($\bar{r} \approx 0.012$ vs. 0.006), but the bump is gone by $k = 30$. Extremely template-aligned targets matter only at the very tail.

2.4 Chapter 7: Template alignment vs. pairwise cohesion

Plain-English intent: Now the same idea per-stimulus: does each target's *individual* template alignment predict how similar it is to the other targets? A simple scatter and OLS line.

For each target i, plot $r(s_i, t_{\text{wall}})$ against $\bar{r}(s_i, S_{\text{tgt}\setminus i})$. OLS regression gives:

$$\text{slope} = 0.0342, \qquad R = 0.119, \qquad R^2 = 0.014, \qquad p = 0.148.$$

Template alignment explains $\sim$1.4% of the variation in how-much-each-target-resembles-the-others. The relationship is positive but extremely weak.

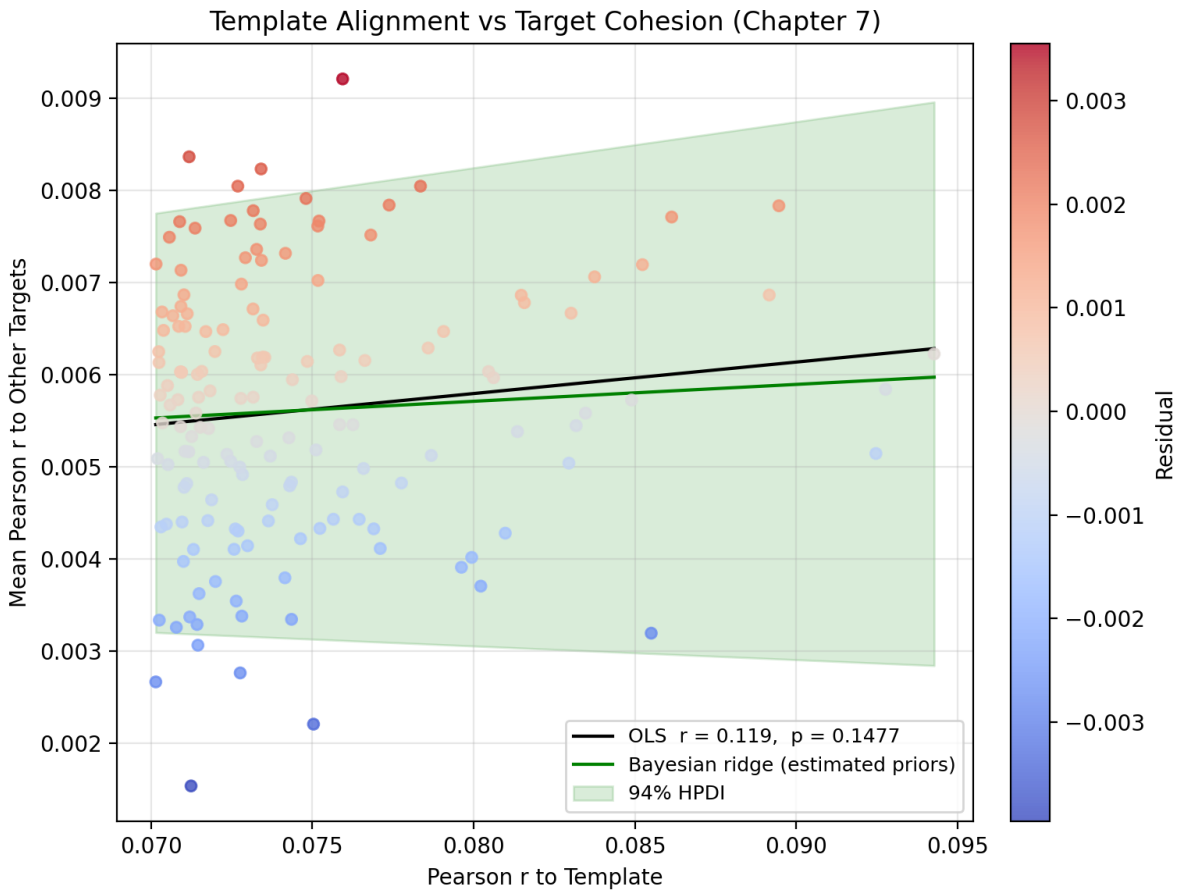


Figure 2: Per-target template alignment vs. mean similarity to other targets.

Reader takeaway: The relationship is positive but barely there ($R = 0.12$, $R^2 = 1.4\%$, $p = 0.15$). Template alignment is a small part of why a target resembles other targets; most variance in per-target cohesion is something else (or noise).

2.5 Chapter 8: Pearson-based geometry of the full set

Plain-English intent: If we take the 300×300 Pearson similarity as a proxy for stimulus geometry, what does that geometry look like? We embed (MDS, PCA, t-SNE), cluster blindly, and separately ask whether the experimenter's target–distractor mean-difference direction is a statistically privileged axis even when no clustering recovers it.

Build the symmetric distance $D = \mathrm{clip}(1 - R, 0, \cdot)$ from the full 300×300 Pearson similarity, and embed.

Classical MDS. Double-centre the squared-distance matrix to form the Gram matrix

$$B \;=\; -\tfrac{1}{2}\, J\,(D \odot D)\, J, \qquad J = I - \tfrac{1}{n}\mathbf{1}\mathbf{1}^{\top},$$

take eigendecomposition $B = U \Lambda U^{\top}$, and use the top-k positive eigenvectors scaled by $\sqrt{\Lambda_+}$ as coordinates.

Linear PCA on raw waveforms for comparison.

t-SNE on the precomputed Pearson distance.

Clustering with agglomerative average-linkage and spectral k-means, $k \in \{2, 3, 4, 5\}$, scored by silhouette and Adjusted Rand Index (ARI) against experimenter labels.

Label axis. Define the unit direction

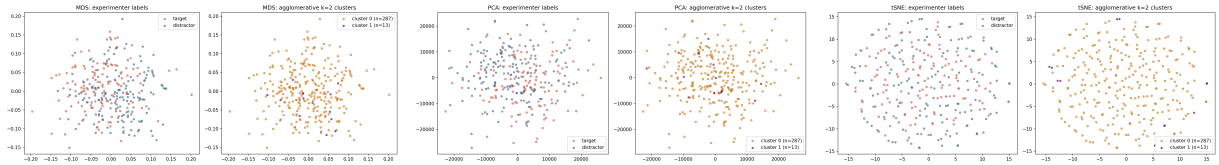
$$\hat{u} \;=\; \frac{\bar{s}_{\mathrm{tgt}} - \bar{s}_{\mathrm{dist}}}{\|\bar{s}_{\mathrm{tgt}} - \bar{s}_{\mathrm{dist}}\|},$$

project every clip onto $\hat{u}$, and compute Fisher's ratio $F = (\mu_t - \mu_d)^2/(\sigma_t^2 + \sigma_d^2)$ and the variance fraction $\mathrm{Var}(s \cdot \hat{u}) \,/\, \sum_j \mathrm{Var}(s_{\cdot j})$. Compare both against 2,000 random unit directions.

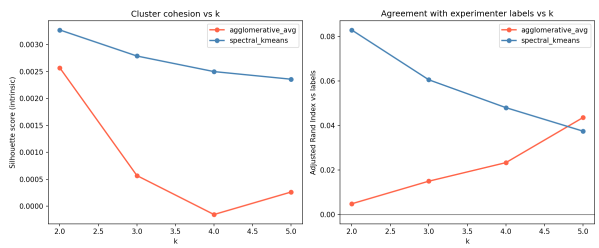
Table 3: Chapter 8 results.

Quantity	Observed	Null mean	p
MDS top-2 variance fraction	1.63%	—	—
PCA top-2 variance fraction	1.11%	—	—
Best silhouette ($k = 2$, spectral)	0.0033	—	—
Best ARI vs. labels ($k = 2$, spectral)	0.083	—	—
Fisher ratio along label axis	37.67	0.009	$< 10^{-3}$
Variance fraction along label axis	0.493%	0.034%	$< 10^{-3}$

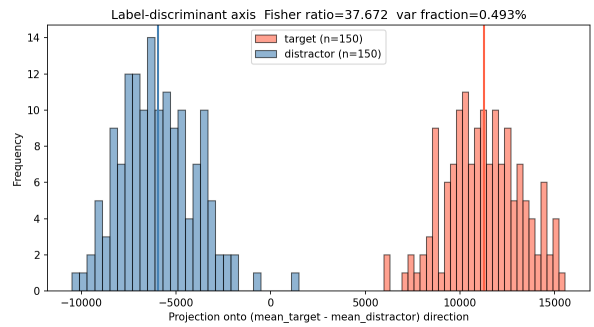
Interpretation. The set has no low-dimensional shape: the leading two MDS components explain only 1.6% of Gram variance. Generic clustering finds nothing aligned with labels (best ARI $= 0.08$). But the target–distractor mean-difference direction is a statistically privileged axis ($\sim$37$\times$ stronger than random) that simply isn't dominant enough ($\sim$0.5% of waveform variance) to organise clusters. Targets are biased in a single shared direction (toward `wall.wav`) without being internally tight.



(a) Classical MDS (b) PCA (c) t-SNE



(d) Silhouette and ARI vs. k



(e) Projection onto the label axis $\hat{u}$

Figure 3: Chapter 8 outputs.

Reader takeaway: Two facts you have to hold in your head simultaneously: (a) generic blind clustering finds nothing label-related (best ARI 0.08, best silhouette 0.003), but (b) the explicit label-axis direction $\hat{u}$ is *statistically privileged* ($F = 37.7$, $p < 10^{-3}$, $\sim 37\times$ stronger than a random direction). The label is a real but extremely low-variance direction (0.49% of waveform variance) — too weak to drive clusters but too systematic to be noise.

2.6 Chapter 9: Template projection and residual structure

Plain-English intent: Why does the target–target Pearson cohesion exist at all? Pairwise Pearson decomposes *exactly* into a contribution from the template projection $\alpha_i t$ plus a contribution from the residuals $e_i = s_i - \alpha_i t$. We compute both and see which one carries the cohesion.

2.6.1 Decomposition

Let $t = t_{\text{wall}}$ centred, and s_i centred clip i. The OLS projection is

$$\alpha_i \;=\; \frac{\langle s_i, t\rangle}{\langle t, t\rangle}, \qquad \hat{s}_i = \alpha_i\, t, \qquad e_i = s_i - \hat{s}_i. \tag{2}$$

Pairwise Pearson decomposes exactly:

$$\langle s_i, s_j\rangle \;=\; \alpha_i\alpha_j\langle t, t\rangle \;+\; \langle e_i, e_j\rangle, \tag{3}$$

and dividing by $\sigma_{s_i}\sigma_{s_j}$ gives

$$r(s_i, s_j) \;=\; \underbrace{\frac{\alpha_i\alpha_j\,\|t\|^2}{\sigma_{s_i}\sigma_{s_j}}}_{\text{template contribution}} \;+\; \underbrace{\frac{\langle e_i, e_j\rangle}{\sigma_{s_i}\sigma_{s_j}}}_{\text{residual contribution}}. \tag{4}$$

Listing 2: Decomposition core (`ch9-projection-residual.py`).

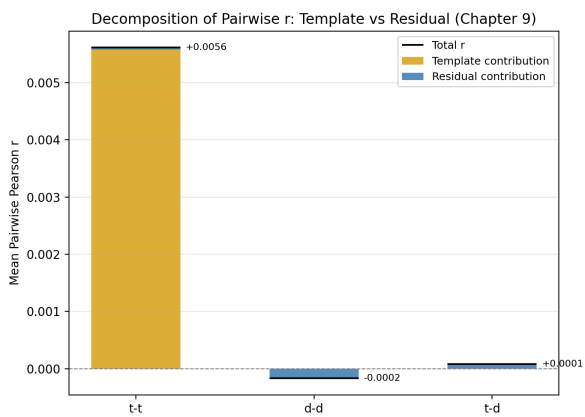
```
alpha = (X @ t) / (t @ t) # (n,)
proj = np.outer(alpha, t) # (n, L)
resid = X - proj # (n, L)
sigma = X.std(axis = 1, ddof = 0) # (n,)
tmpl_contrib = np.outer(alpha, alpha) * (t @ t) / np.outer(sigma, sigma)
resid_contrib = (resid @ resid.T) / L / np.outer(sigma, sigma)
# r_total[i,j] == tmpl_contrib[i,j] + resid_contrib[i,j] (verified <= 1.4e-16)
```

2.6.2 Mean contributions

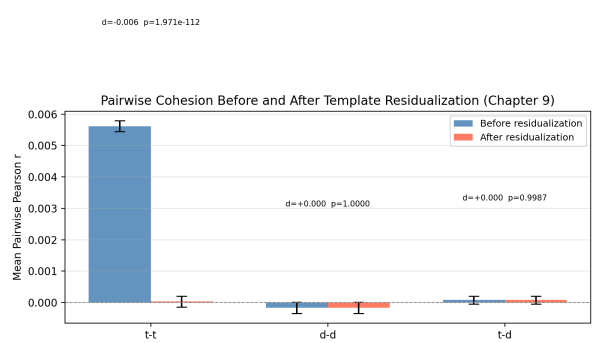
Table 4: Decomposition of mean pairwise Pearson into template and residual parts.

Pool	$\overline{r}$	template part	residual part	frac. template
target–target	$+5.61\times10^{-3}$	$+5.58\times10^{-3}$	$+3.5\times10^{-5}$	99.4%
distractor–distractor	-1.67×10^{-4}	~ 0	-1.67×10^{-4}	~ 0
target–distractor	$+8.1\times10^{-5}$	~ 0	$+8.1\times10^{-5}$	~ 0

After residualising the template, target–target cohesion is no longer distinguishable from zero ($t = 0.20$, $p = 0.84$, Wilcoxon $p = 0.57$, Welch before-vs-after $p = 2 \times 10^{-112}$). The targets cohere *entirely* because they each contain a small piece of `wall.wav`; once removed, they revert to noise.



(a) Total $\bar{r}$ split into template + residual contributions.



(b) Before/after residualisation, by group.

2.6.3 Per-sample contributions (Chapter 9b)

The pairwise contribution at sample ℓ is $c_\ell^{(ij)} = s_{i,\ell}\, s_{j,\ell}$. Averaging $c_\ell^{(ij)}$ over all target–target pairs gives a per-sample profile of *which timepoints carry the target–target similarity.* Correlating that profile against t_ℓ^2 (template energy), $|t_\ell|$, and t_ℓ (signed):

Table 5: Per-sample target–target contribution profile vs. template features.

Comparison	r	p
$\overline{c_\ell^{(tt)}}$ vs. t_ℓ^2	+0.623	$\approx 10^{-317}$
$\overline{c_\ell^{(tt)}}$ vs. $\lvert t_\ell \rvert$	+0.575	$\approx 10^{-260}$
$\overline{c_\ell^{(tt)}}$ vs. t_ℓ	+0.224	$\approx 10^{-35}$
$\overline{c_\ell^{(dd)}}$ vs. t_ℓ^2	+0.007	0.70

The samples that carry target–target similarity are the same samples where the template is loud. Distractor pairs show no such pattern.

Reader takeaway: 99.4% of the target–target mean Pearson is the template contribution $\alpha_i\alpha_j\|t\|^2/(\sigma_i\sigma_j)$. Once that is subtracted, target–target cohesion is statistically zero ($p = 0.84$). Targets are *not* a coherent acoustic family; they are a set of clips that each contain a small piece of `wall.wav`.

2.7 Chapter 10: Null models for Pearson structure

Plain-English intent: The target–target excess is small ($\Delta\bar{r} \approx 0.006$). Is even that meaningful? Two complementary nulls: (a) replace all 300 clips with fresh white noise and ask whether $\bar{r}_{tt}$ this big could happen by chance (white-noise null); (b) keep the data fixed but shuffle the labels and recompute (label-permutation null).

Two complementary nulls.

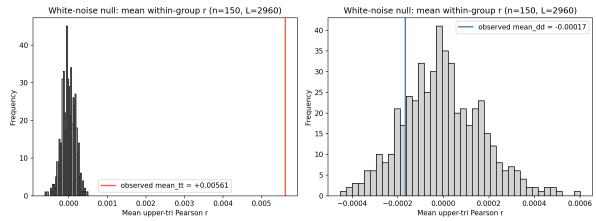
White-noise null. Synthesise 300 clips of length L as iid Gaussian noise, compute their pairwise Pearson, take mean within-set. Repeat 500 times.

Label-permutation null. Hold the actual 300×300 matrix fixed, permute the target/distractor labels, recompute mean within-tt and within-dd. Repeat 5,000 times.

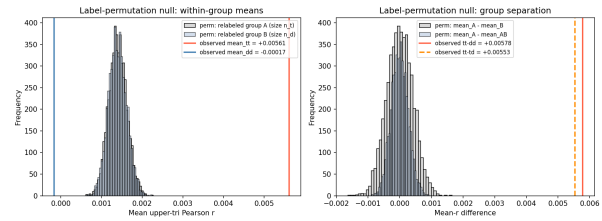
Table 6: Null-model summary (Chapter 10).

Test	observed	null mean	null std	p
white-noise null, $\bar{r}_{tt}$	+0.00561	~ 0	1.7×10^{-4}	0.002
white-noise null, $\bar{r}_{dd}$	−0.00017	~ 0	1.7×10^{-4}	0.33
label-perm, $\bar{r}_{tt}$	+0.00561	+0.00140	2.2×10^{-4}	$< 2\times10^{-4}$
label-perm, $\bar{r}_{tt} - \bar{r}_{dd}$	+0.00578	$+6\times10^{-6}$	4.1×10^{-4}	$< 2\times10^{-4}$

The target–target cohesion sits at the 100th percentile of both nulls; distractors are indistinguishable from white noise (which is exactly what the selection rule tried to achieve).



(a) White-noise null.



(b) Label-permutation null.

Reader takeaway: The observed cohesion sits at the 100th percentile of both nulls ($p = 0.002$ white-noise, $p < 2 \times 10^{-4}$ label-permutation). The distractor pool is indistinguishable from white noise, exactly as the selection rule intended.

3 Non-Pearson metric analyses (Part III)

3.1 Chapter 11: Choosing alternative metrics

Plain-English intent: The selection rule was Pearson on raw centred waveforms, but a human listener perceives audio through filters that look very little like that. We define four perceptually-motivated similarity measures (envelope-based, spectral, cepstral, time-shift-tolerant) so that subsequent chapters can ask the label question through several different lenses.

Five similarity measures, each chosen for an interpretable axis of acoustic structure.

1. Pearson raw (reference). $r(s_i, s_j) = \cos(s_i, s_j)$ on centred waveforms.

2. Lagged cross-correlation, ± 200 ms. Full sample-domain cross-correlation via FFT: $x_{ij}(\tau) = \sum_t s_{i,t} s_{j,t+\tau} \,/\, (\|s_i\| \|s_j\|)$, report $\max_{|\tau| \le \tau_{\max}} |x_{ij}(\tau)|$. Captures similarity that survives small temporal shifts.

3. Envelope correlation. Compute the analytic-signal magnitude (Hilbert envelope), low-pass with a 161-sample moving average (~ 50 Hz cutoff at 8 kHz), then Pearson on envelopes. Removes carrier phase, keeps loudness contour.

4. Log-spectrum correlation. $\tilde{s} = \log(1 + |\mathcal{F}(s)|)$ from the real FFT, then Pearson across frequency bins. Compares spectral shape, ignores phase.

5. MFCC cosine. 12 Mel-frequency cepstral coefficients per frame (dropping C_0, which collapses to a constant under RMS normalisation), summarised by mean and std across frames, z-scored per dimension across clips, cosine similarity. Approximates a coarse speech-style timbral descriptor.

Listing 3: Lagged cross-correlation core.

```
F = next_fast_len(2*L - 1)
S = np.fft.rfft(X, n = F, axis = 1) # (n, F/2+1)
norms = np.linalg.norm(X, axis = 1)
for i in range(n):
    cc = np.fft.irfft(S[i] * np.conj(S), n = F, axis = 1)
    cc = np.concatenate([cc[:, -max_lag:], cc[:, :max_lag+1]], axis = 1)
    out[i, :] = np.max(np.abs(cc), axis = 1) / (norms[i] * norms)
```

Reader takeaway: Five orthogonal lenses on the same 300 clips. Pearson is the selection metric; the other four ask whether the label structure shows up under time-shift-tolerant, envelope, spectral, or cepstral similarity.

3.2 Chapter 12: Label-aware separation under each metric

Plain-English intent: Repeat the chapter 5 target–target / distractor–distractor split for each non-Pearson metric. If any of these acoustic descriptions agreed with the label, we would see a non-trivial Cohen's d.

Same target–target / distractor–distractor / target–distractor decomposition as Chapter 5, repeated for each metric.

Table 7: Cohen's d for tt vs. dd under each metric.

Metric	$\bar{r}_{tt}$	$\bar{r}_{dd}$	d_{tt-dd}
pearson_raw	+0.00561	−0.00017	**0.313**
lagged_xcorr_max	+0.06186	+0.06165	0.037
envelope_corr	+0.74480	+0.74344	0.022
log_spectrum_corr	+0.17737	+0.17726	0.005
mfcc_cosine	−0.00072	−0.00137	0.003

Interpretation. Pearson is the only metric under which the target/distractor split is visible at all. Under envelope correlation the average pair has $r \approx 0.74$ regardless of label — all clips look alike to the envelope, because they all share the same RMS-normalised amplitude contour profile. The label is a property of the metric, not of the stimuli.

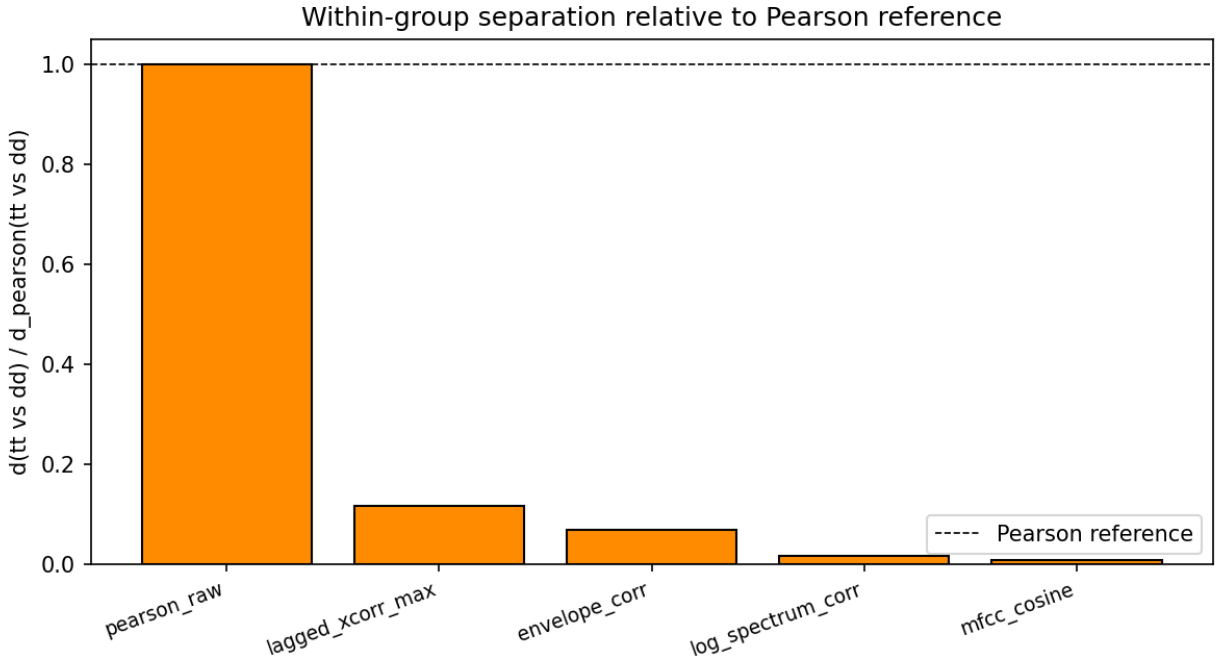


Figure 6: Per-metric label separation, normalised against Pearson.

Reader takeaway: Pearson is the only metric under which the label is even faintly visible ($d = 0.31$). Lagged xcorr, envelope, log-spectrum, MFCC all give $d \leq 0.04$. The target/distractor distinction is a Pearson artefact, not an acoustic property a listener could realistically pick up.

3.3 Chapter 13: Label-blind clustering under each metric

Plain-English intent: Forget the labels and ask: under each metric, does the set break into natural clusters at $k = 2$? If acoustic structure exists that is not the experimenter's split, blind clustering should find it.

Convert similarity $s_{ij} \in [-1, 1]$ to distance $d_{ij} = \text{clip}(1 - s_{ij}, 0, 2)$ and to affinity $a_{ij} = (s_{ij} + 1)/2$. Cluster at $k = 2$ with

- agglomerative average linkage on D,
- spectral k-means on A,

and score with silhouette (intrinsic) and ARI vs. experimenter labels.

Table 8: Chapter 13 results, $k = 2$.

Metric	Method	sizes	silhouette	ARI
pearson_raw	agglomerative	$[287, 13]$	0.003	0.005
pearson_raw	spectral	$[156, 144]$	0.003	0.083
lagged_xcorr_max	agglomerative	$[256, 44]$	0.001	−0.001
lagged_xcorr_max	spectral	$[166, 134]$	0.001	0.022
envelope_corr	agglomerative	$[294, 6]$	**0.151**	−0.000
envelope_corr	spectral	$[118, 182]$	0.087	−0.003
log_spectrum_corr	agglomerative	$[298, 2]$	0.011	−0.000
log_spectrum_corr	spectral	$[145, 155]$	0.005	0.000
mfcc_cosine	agglomerative	$[203, 97]$	0.055	−0.001
mfcc_cosine	spectral	$[154, 146]$	0.086	−0.003

The best silhouette (0.15, envelope agglomerative) corresponds to a 294/6 split, i.e. peeling six acoustic outliers off the rest. No clustering recovers the experimenter labels (max ARI 0.083).

Reader takeaway: The best cluster split (envelope agglomerative, silhouette 0.15) peels off six acoustic outliers from the other 294. There is no clean $k = 2$ partition under any metric, and certainly none aligned with the experimenter's labels.

3.4 Chapter 14: Cross-metric synthesis

Plain-English intent: Bring the five metrics onto a single scoreboard along three axes (target cohesion, label separation, intrinsic clustering), and check whether the metrics describe redundant geometries (Mantel-style cross-metric correlations).

Three axes summarise each metric:

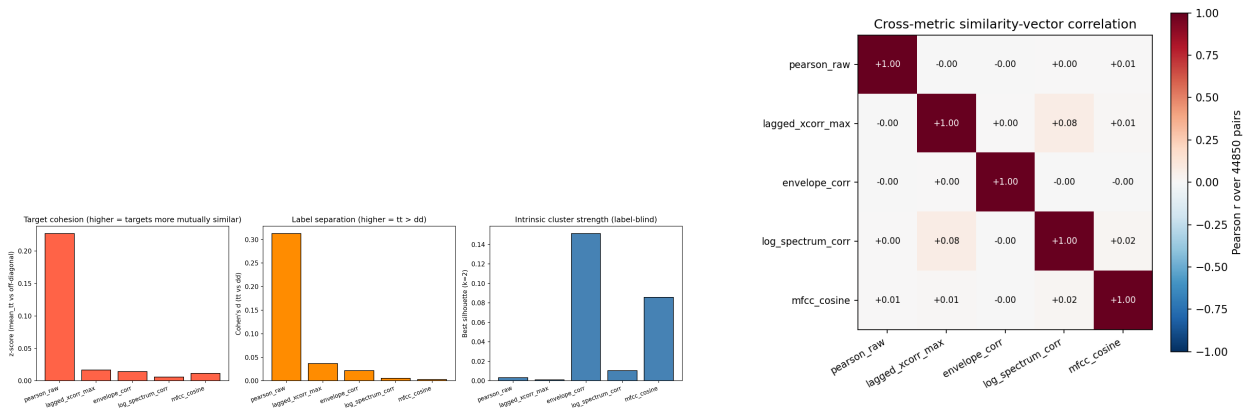
1. **Target cohesion** (z-score of $\bar{r}_{tt}$ against the metric-specific off-diagonal distribution),
2. **Label separation** (Cohen's d_{tt-dd}),

3. **Intrinsic clustering** (best silhouette over methods at $k = 2$).

Table 9: Cross-metric ranking.

Metric	cohesion (z)	label sep. (d)	intrinsic silh.
pearson_raw	**0.227**	**0.313**	0.003
lagged_xcorr_max	0.017	0.037	0.001
envelope_corr	0.014	0.022	**0.151**
log_spectrum_corr	0.006	0.005	0.011
mfcc_cosine	0.012	0.003	0.086

Mantel-style cross-metric similarity. Flatten each metric's upper-triangular pair vector and compute Pearson's r between metrics. Maximum off-diagonal $|r| = 0.078$ (lagged vs. log-spectrum), most $|r| < 0.01$. The five metrics are nearly orthogonal as ways of describing this set. No metric is a disguised duplicate of another.



(a) Per-metric rankings on the three axes.

(b) Cross-metric Mantel correlations.

Reader takeaway: The five metrics are nearly orthogonal as descriptions of this set (max cross-metric $|r| = 0.078$). No metric is a redundant copy of another; each really is asking a different question. The label is visible only to Pearson; intrinsic clustering structure is visible only to envelope.

4 Synthesis and behavioural cross-check (Part IV)

4.1 Chapter 15: What did subjects do?

Plain-English intent: Even though Book 1 is about the stimulus set, we owe ourselves a one-pass behavioural cross-check: do any of these metrics relate to subject responses? If not, that is itself the most informative possible result. Full subject-side analysis lives in `book2_writeup.pdf`.

The Book 1 mandate is stimulus-only, but a one-pass behavioural sanity-check answers the question *"does any of this matter for how subjects responded?"* without committing to a full Book 2.

Trials. 26 subjects across two sites, 7,748 trials total ($\sim 2 \times 150$ trials per subject). Each trial yields `(stimulus_number, true_label, subject_response)`.

Per-stimulus feature. For each metric M and each clip i in block B,

$$\Delta_i^M \;=\; \overline{M(s_i, s_j)}_{j \in B,\, \mathrm{label}_j = \mathrm{tgt},\, j \neq i} \;-\; \overline{M(s_i, s_j)}_{j \in B,\, \mathrm{label}_j = \mathrm{dist},\, j \neq i}.$$

A clip with large positive Δ^M is more like targets than distractors under M.

Stimulus-level prediction. Pearson r between Δ_i^M and the empirical $p_i = \Pr(\text{subject called } s_i \text{ a target})$:

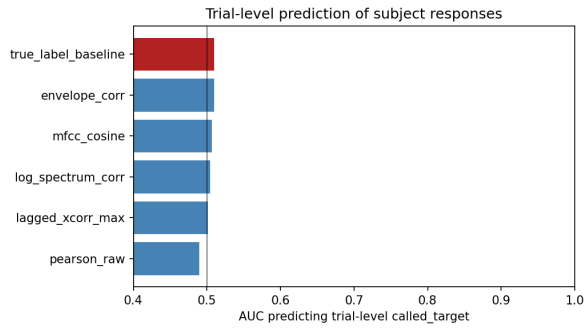
Metric	r	p
pearson_raw	+0.094	0.11
envelope_corr	+0.078	0.18
mfcc_cosine	+0.059	0.31
log_spectrum_corr	+0.044	0.45
lagged_xcorr_max	−0.007	0.91

Trial-level prediction (logistic AUC). Predict Pr(called target) from Δ^M alone, vs. from the experimenter's true label:

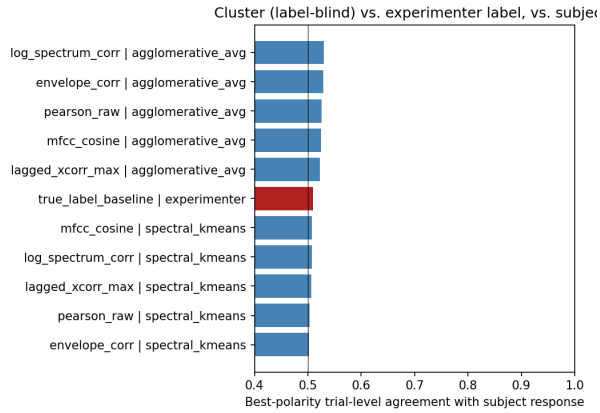
Predictor	AUC
true_label_baseline	0.510
envelope_corr	0.510
mfcc_cosine	0.507
log_spectrum_corr	0.504
lagged_xcorr_max	0.502
pearson_raw	0.490

Cluster vs. response agreement. Match each metric's label-blind $k = 2$ clustering against the binary subject response, picking the better of the two polarities:

Metric	method	match rate
log_spectrum_corr	agglomerative	**0.530**
envelope_corr	agglomerative	0.529
pearson_raw	agglomerative	0.526
mfcc_cosine	agglomerative	0.524
lagged_xcorr_max	agglomerative	0.522
true_label_baseline	—	0.510



(a) Trial-level AUC.



(b) Cluster-vs-response agreement.

Interpretation.

1. Subjects performed at chance. Mean accuracy 0.510, hit rate 0.480, false-alarm rate 0.460.
2. No metric, including the experimenter labels, predicts trial responses meaningfully.
3. The label-blind agglomerative clusters consistently beat the experimenter labels by ~ 2 percentage points. These clusters tend to isolate small acoustic-outlier groups, suggesting subjects were weakly tracking gross acoustic salience rather than the imposed identity contrast.

Reader takeaway: Subjects were at chance (mean accuracy 0.510). No metric — not even the experimenter labels — predicts trial responses meaningfully (best AUC 0.510). Surprisingly, blind acoustic clusters fit subject responses *better* than the experimenter labels, by about 2 points. This motivated Book 2.

5 What the stimulus set actually contains

Pulling it all together:

Structure imposed by design. A small ($\Delta\bar{r} \approx 0.006$, $d \approx 0.31$) but statistically robust target–distractor separation under Pearson, driven *entirely* by shared projection onto `wall.wav`.

Structure actually present. Beyond the wall-projection direction, essentially noise. Targets are not internally cohesive once the template is removed. No metric reveals interesting intrinsic clusters; the set is a 300-point cloud in a high-dimensional space with one weak preferred direction.

Structure visible only under some metrics. Only Pearson sees the label. Under any other reasonable acoustic similarity (envelope, lagged r, log-spectrum, MFCC), targets and distractors are statistically indistinguishable. Subjects, who presumably do not perform sample-level Pearson against the template in their heads, would have no acoustic basis to recover the labels.

Implications for subject analyses. Behavioural performance was at chance, consistent with the conclusion above. The interesting variance for downstream books is between *subjects*, not between stimuli. Per-stimulus prediction of behaviour by acoustic features is a dead end with this stimulus set.

A File map

Scripts (all under `stimuli-analyses/`).

- `pairwise-pearson.py` — Chapter 5
- `target-cohesion-via-rank.py` — Chapters 6, 7
- `ch8-pearson-geometry.py` — Chapter 8
- `ch9-projection-residual.py` — Chapter 9
- `ch9b-per-sample-contributions.py` — Chapter 9b
- `ch10-null-models.py` — Chapter 10
- `ch11-non-pearson-metrics.py` — Chapter 11 (caches metrics)
- `ch12-label-aware-non-pearson.py` — Chapter 12
- `ch13-label-blind-non-pearson.py` — Chapter 13
- `ch14-metric-comparison.py` — Chapter 14
- `ch15-subject-response-prediction.py` — Chapter 15

Outputs. Tables and figures live under `stimuli-analyses/outputs/{pearson,non_pearson}/<chapter>/`.

B Key formulae cheat sheet

$$
\begin{aligned}
r(s,t) &= \frac{\langle s_c, t_c\rangle}{\|s_c\|\,\|t_c\|} \qquad \text{(centred Pearson)}\\
\alpha_i &= \frac{\langle s_i, t\rangle}{\langle t, t\rangle} \qquad \text{(OLS template projection)}\\
e_i &= s_i - \alpha_i t \qquad \text{(residual)}\\
r(s_i, s_j) &= \frac{\alpha_i\alpha_j\|t\|^2 + \langle e_i, e_j\rangle}{\sigma_{s_i}\sigma_{s_j}}\\
F &= \frac{(\mu_t - \mu_d)^2}{\sigma_t^2 + \sigma_d^2} \qquad \text{(Fisher discriminant ratio)}\\
\hat{u} &= \frac{\bar{s}_{\text{tgt}} - \bar{s}_{\text{dist}}}{\|\bar{s}_{\text{tgt}} - \bar{s}_{\text{dist}}\|} \qquad \text{(label axis)}\\
B &= -\tfrac{1}{2}\,J\,(D \odot D)\,J, \quad J = I - n^{-1}\mathbf{1}\mathbf{1}^\top \qquad \text{(classical MDS Gram)}\\
d &= \frac{\mu_a - \mu_b}{\sqrt{\frac{1}{2}(\sigma_a^2 + \sigma_b^2)}} \qquad \text{(Cohen's } d\text{)}.
\end{aligned}
$$